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A Demonstration of the Benefits of Aggregation in an Analytic Task

Described herein is an easily done exercise that demonstrates the benefits of aggregated analysis. It might be employed by instructors of intelligence analysis as an exercise in the classroom, or by the intelligence practitioners seeking some ''hands on'' experience with something they have heard of, but have not personally worked with.

Aggregated analysis refers to the taking of a large number of independent assessments and mathematically aggregating them to form a single assessment. This process does not include brainstorming, for example, where individuals actively interact with each other while forming the group assessment.[2]

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Average assessments derived from a large number of independent estimates tend to be quite accurate. Research on this topic was popularized by James Surowiecki in his book *The Wisdom of Crowds*.[3] The archetypal story is told about Sir Francis Galton. While attending the West of England Fat Stock and Poultry Exhibition, Galton noted a contest where competitors guessed the weight after butchering of a fat ox. Galton collected the almost 800 guesses transcribed on cards and discovered that the mean estimate (1,197 lbs.) was nearly identical to the actual weight (1,198 lbs.) of the butchered ox.[4]

Researchers have examined the aggregation question more rigorously over the years. Aggregating the independent assessments of a large number of individuals has been observed to provide benefits in a wide variety of disciplines. A sampling: executives at *Time* magazine made forecasts on the amount of advertising space they would sell in future issues. With ad sales being a primary source of revenue in many publications, such forecasts receive a lot of attention and are given great importance. Individual forecasters had a forecasting error of 10.5 percent. If two individual forecasters had their predictions averaged, the subsequent error rate fell to 8.9 percent. Error rates continued to fall as the assessments of additional forecasters were added to the mix. A group of thirteen forecasters reduced the average error rate to 7.2 percent.[5] In another example, airlines consistently attempt to forecast how many passengers will fly in the upcoming months. Such forecasts are important so that airlines can have enough aircraft and other resources where needed. In one study, two separate sets of forecasts, when combined, provided a more accurate estimate than either set did alone.[6] One way to efficiently aggregate a large number of independent forecasts is to utilize a stock market mechanism. Futures markets use this principle to evaluate the likelihood of future events. Forecasts based on futures markets have been found highly accurate in assessments of such diverse matters as movie box-office earnings, mobile phone usage, and even pop-music chart positions.[7]

Efforts to apply the methodology continue. Among them is the Intelligence Advanced Research Projects Activity's (IARPA) Aggregate Contingent Estimation (ACE) program, whose purpose is to "dramatically enhance the accuracy, precision, and timeliness of intelligence forecasts for a broad range of event types, through the development of advanced techniques that elicit, weight, and combine the judgments of many intelligence analysts."[8] In addition to Intelligence Community (IC) efforts, futures markets that are open to the public (InTrade, HSX, and the Iowa Electronics Market) continue to estimate the likelihood of various events, and are shown to be more accurate than other benchmarks of performance.[9]

And yet, to most in the Intelligence Community, talk of these topics is esoteric. Few are likely to anticipate starting their own complicated

Bayesian aggregation algorithms to combine forecasts (as is done in the IARPA program) or setting up their own political prediction market. The advantages of crowd-based analysis seem to be merely an academic discussion. What analysts believe is needed is a more accessible, easily demonstrable example of the crowd tackling an analytic task. Such an example can serve as an introduction to the benefits of aggregated analysis as well as a demonstration to skeptical intelligence professionals who want to evaluate a technique first-hand before making up their own minds.

IN SEARCH OF AN ANALYTIC TASK

The most suitable intelligence demonstration for an analytic task is an assessment of some issue that is relevant to national security. Will Iran develop a nuclear weapon in 2015? Will Israel conduct an air strike against Iran in the next six months? Will North Korea conduct another test of a nuclear weapon in the next three months? Will Assad retain power in Syria? Questions such as these are asked in IARPA's ACE program, as well as prediction markets such as InTrade.[10] But such questions unfold over a matter of months, or even years, and are therefore less suitable to serve as a quick introductory demonstration on the effectiveness of aggregated analysis. Students are in the classroom for a limited length of time, and intelligence professionals may be too busy to invest in such a long evaluation period to acquire first-hand experience with a method. Needed instead is a short, quick, and easily demonstrable analytic task.

Attempting to fill this void, the following admittedly overly simplistic analogy of an analytic task is offered to serve as a suitable demonstration. Though lacking some of the complexity of a more realistic intelligence analytic task, it offers some similarities and is brief enough to be demonstrated from start to finish in less than an hour. Imagine watching a video clip of a person's face. The person looks, and smiles. The smile might be a genuine reaction to seeing something funny, or it may be a forced smile. What's the difference? The importance of the test can be exaggerated by telling students that if they ever hoped to become quality human intelligence (HUMINT) officers, they'll need to judge the character of a potential source by looking the person in the eye and determining his/her authenticity immediately. The "analyst" in this case must make a subjective judgment, by looking for any number of cues to indicate the smile's validity. But, as in real world intelligence analysis, describing all those cues, or describing the overall certainty surrounding the assessment, may be difficult. (This test can be accessed by going to the footnoted Web address.[11]) Can the benefits of aggregated analysis be demonstrated in such a task?

I asked 217 people to take this test. Each person assessed a total of 20 separate videos of individuals smiling. On average, most people didn't do particularly well. The average individual performance was 13.6 out of 20 smiles judged correctly. I also attempted to find individual characteristics that would identify those people who would do best at this task. I examined gender, age, number of children (what better training to detect deception than to have a child!), as well as the number of years someone was in a profession where "reading" people was required, such as investigator, market researcher, survey administrator, polygrapher, or other similar position. None of these characteristics was a significant predictor of performance. The best performance was achieved by three individuals who were correct in 19 out of 20 of their assessments. The video judged incorrectly was unique for each of these top performing individuals. All got one judgment incorrect, but they did not get the same one incorrect. (See Figure 1.)

The data shows something else too. Looking at just the first smile video by itself, most students (52 percent) thought the smile was genuine. Most were also correct. The smile was genuine. Most students (87 percent) thought the second smile was forced. This was also correct. In fact, in going through each of the 20 videos in this manner, the prevailing opinion was correct in 19 out of 20 smile videos, for an impressive accuracy rate of 95 percent. This, in spite of the fact that individual performance averaged only 68 percent. The group performance succeeded grandly where individual performance was mediocre. No individual did better than the group performance.

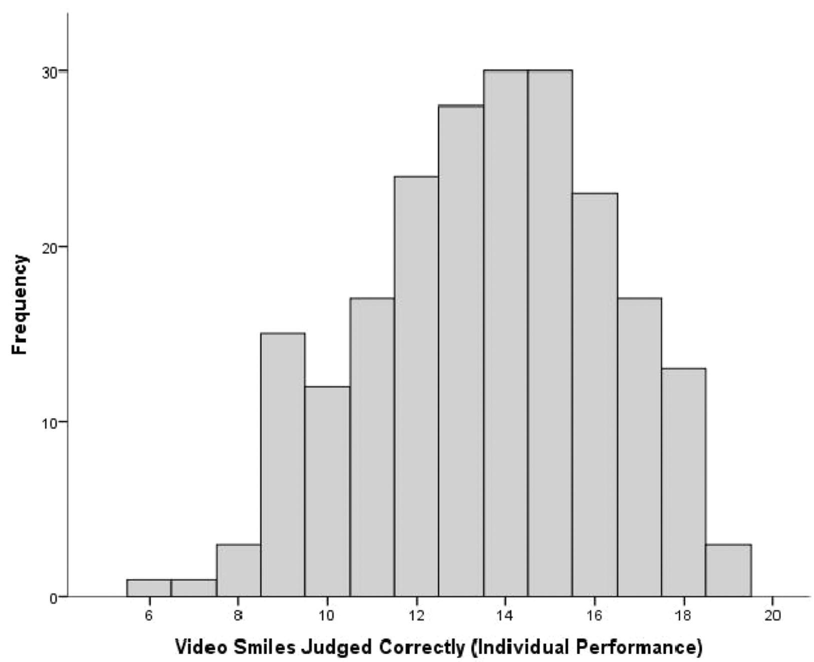

Figure 1. Histogram showing individual performance on the "Smile Video Task".

TASK PERFORMANCE FOR DIFFERENT SIZED GROUPS

A group of 217 is certainly larger than the typical academic class size which might be used to demonstrate aggregated analysis. How would the performance level of a typical college class of around 21 students perform? To simulate how a hypothetical group would perform on the smile video task, 21 individuals were selected at random from the 217 that had previously performed the task. A virtual vote was held to determine the class assessment for each of the twenty smile videos. For example: on the first smile video, if 11 class members deemed the smile video to be genuine, and 10 assessed it as forced, the assessment for the first video would then be that it was a genuine smile. This was done for each of the twenty smile videos. Finally, the group's assessments were graded to see how many videos they got correct.

The individuals making up the first hypothetical class of 21 were then returned to the larger pool of individuals, and the whole procedure was repeated to create a second hypothetical class of 21. By repeating this procedure multiple times, and obtaining a large number of hypothetical classes, a more accurate estimate of what constitutes "typical" performance by a group of 21 was obtained. In all, 217 hypothetical classes of 21 individuals were created and graded. On average, a group of 21 individuals scored 86 percent (17.2 out of 20 correct) on the smile video task, a significant increase over the typical individual performance of 68 percent (13.6 out of 20 correct).[12] (See Figure 2.)

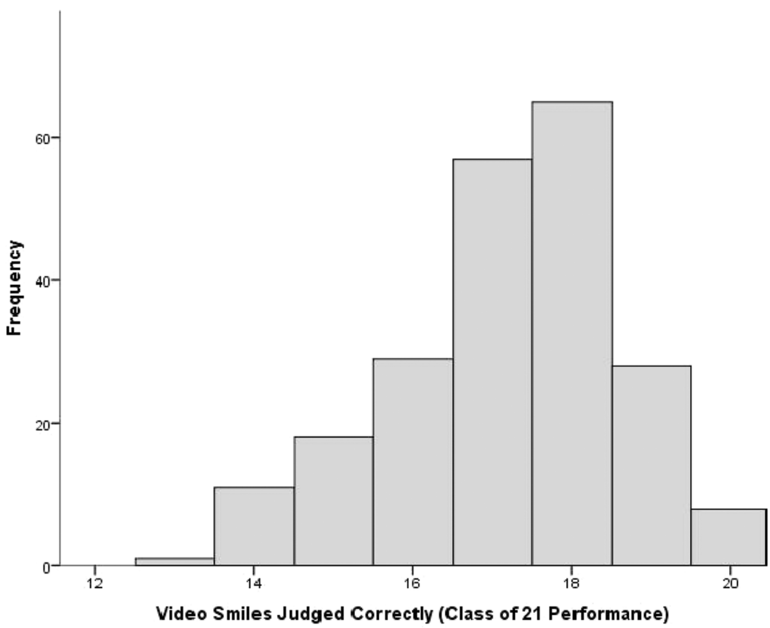

Figure 2. Histogram showing aggregate analysis of notional group of 21 performance.

What results can be expected if this exercise is utilized in other classes? Assuming a hypothetical class size of 21 individuals, and based on the observed results, some predictions are possible. Most individuals (90 percent) will obtain an individual accuracy rate of between 46 percent (9.2 of 20 judged correctly) and 90 percent (18 of 20 correct).[13] When the individual performance scores in a class of 21 students are averaged, the average individual performance for the class should (90 percent of the time) fall between 63 percent accuracy (12.6 of 20) and 73 percent accuracy (14.6 of 20).[14] Note that this isn't aggregated analysis; it is simply the average individual performance in a class of 21 persons. And how about aggregated analysis? When the entire class independently assesses each smile video, and the assessments are used to create a single class judgment through a voting procedure, as is described in the ''virtual vote'' procedure, most classes (90 percent) who perform this exercise should have an aggregate analytical performance of between 74 percent (14.8 of 20) and 98 percent (19.6 of 20).[15] This is significantly better than the typical average individual performance of between 63 percent and 73 percent.[16]

These results are for a group of 21 persons. How do groups of different sizes fare? I tested different sized groups, starting with a three-person

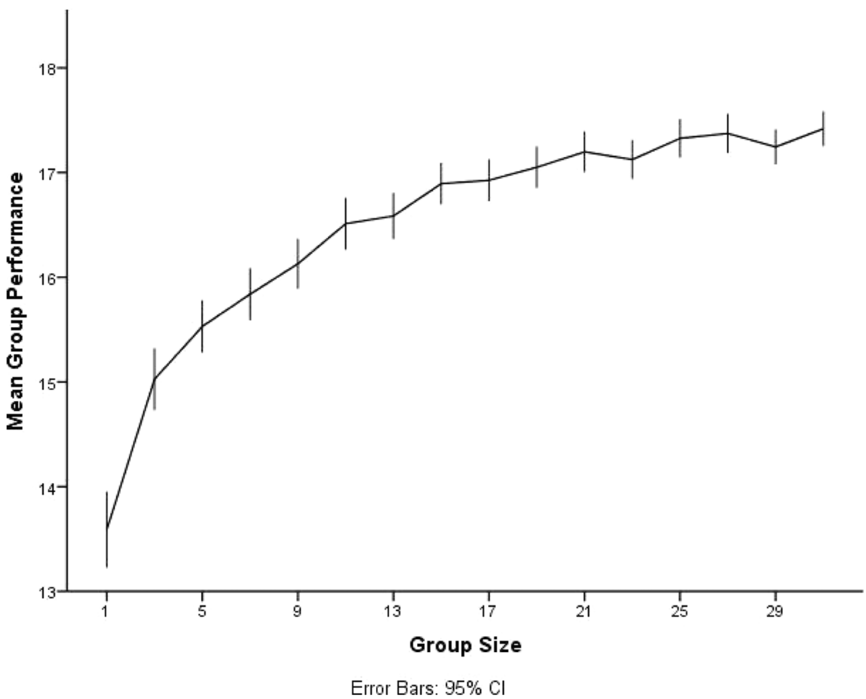

Figure 3. Aggregate analytical performance for groups of various sizes on ''Smile Video Task.''

group and extending all the way up to a 31-person group. For each test of a particular group size, I repeated the same procedures described earlier. Participants were randomly selected from my pool of individual performers and formed into a notional group. A "virtual vote" was conducted to see what the group's assessment would be for each video, and the group's performance was then graded. For each group size tested 217 "virtual groups" were created. As group size increased, the performance of the aggregate analysis continued to improve. For a group-of-three, average performance was 75 percent accuracy.[17] For a group of 31, average performance was 87 percent.[18] (See Figure 3.)

WHAT ABOUT HIGH-PERFORMING INDIVIDUALS?

On occasion, a single individual will equal or exceed the aggregate performance of the group. What should be concluded from this? At least two possibilities exist: skill and luck. The person may have special experience or ability in this task. If this is the case, the individual should be able to deliver a predictably higher performance. This hypothesis can be tested by dividing the smile video test into two parts. Part one would consist of the first 10 videos, part two of the second ten. A person who does extremely well on the first 10 videos, if truly highly skilled, should also do well on the second set. Evidence for this can be found by looking at my results of 217 individuals, where performance on the first 10 videos was significantly correlated to performance on the second ten.[19] This result is consistent with the explanation that an element of skill is present at this task, meaning that high performers are not just lucky.

But, can higher skilled performers ignore the benefits of aggregate analysis? Caution should be exercised in reaching such a conclusion. The aggregate analysis of a group of experts will still outdo the performance of a single expert. In fact, the benefits of aggregate analysis are significant enough that mathematical evidence shows that a large number of laymen can beat a smaller number of experts. For example, if hypothetical laymen make four times as many errors as does an expert, the combined independent assessment of 16 laymen will be needed to equal the performance of one expert. A group of 64 laymen can beat the expert by a factor of two.[20] This is not to say that expertise is not valued in this context. A large number of experts will outperform an equally large number of laymen. Then, again, finding a large number of experts is often difficult.

Then, too, is the matter of consistency. A highly skilled individual should, on average, perform better. Yet, most people have some good days and some bad days. On the bad days, even an expert may return a dismal performance. Performance from an aggregated group tends to be more consistent. The highs are not quite as high, and the lows are not quite as low. The performance range of an aggregated group is more boring and predictable. This can be demonstrated with the data collected in the smile video task. For example: the typical performance of individuals on the smile video task delivered an accuracy rate of between 46 percent and 90 percent—a spread of 44 percentage points.[21] An aggregated group of 21 persons returns a typical accuracy rate of between 74 percent and 98 percent—a noticeably smaller spread of 24 percentage points.[22] The spread for a group of 31 aggregated individuals is even smaller: only 20 percentage points (typically ranging between 77 percent and 97 percent).[23] The range of performance can be more appropriately quantified by calculating the standard deviation. A higher standard deviation indicates a larger "spread" in performance scores. Calculating the standard deviation for groups of different sizes, Figure 4 shows that the "spread" in performance continued to decrease as the group size grew larger.

The results of improved, more predictable performance observed in the smile video task are nothing new. They have been demonstrated and repeated in a host of other studies.[24] But how to explain these results?

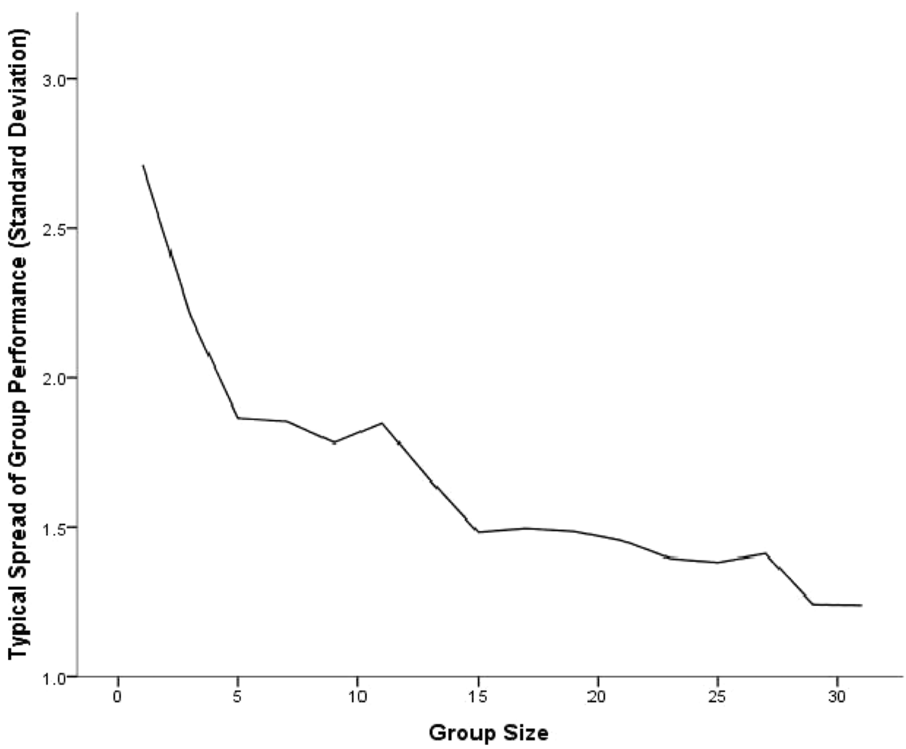

Figure 4. Typical spread of performance for groups of various sizes on "Smile Video Task."

WHY DOES AGGREGATE ANALYSIS DO BETTER?

A few reasons for superior performance of aggregated independent assessments can be highlighted. The first has to do with a mathematical trick related to the central limit theorem. To illustrate, imagine using a tape measure to measure the height of a desk to the nearest millimeter. Then asking another person to do it. And another. And another. At the end of this process, a range of similar measures, but slightly different from each other, would be reported. Of course, the size of the desk wasn't changing. The desk does have a "true" height. But, every time someone measures the desk, a little bit of random error is mixed into the result. Most of the errors in such a test would be quite small. Occasionally an error would be larger. This random error could cause an overestimate of the height of the desk or, equally likely, an underestimate. If the error component is truly random, the chance of overestimating the height is equal to the chance of its being underestimated. Much like flipping a coin, half the coin tosses will be "heads" and half will be "tails." Even though an even split for a small number of coin tosses might not be achieved, as the number of coin tosses grows, the ratio of "heads" and "tails" will be very close to 50/50. Similarly, for a large number of assessments, the random error will be equally spread around the true value. This is an important point because it means that in taking a large number of measures, approximately half of them will be overestimates and the other half will be underestimates. But then the positive errors and negative errors will occur in approximately equal frequency and will cancel each other out when all the measures are aggregated. In the desk case, the average of all the measures is thus likely to be closer to its true height because the random error component has been removed as a result of mutually cancelling errors.

What about non-random error? Systematic error, or bias, is predictable error. If the tape measure used to measure the height of the desk is

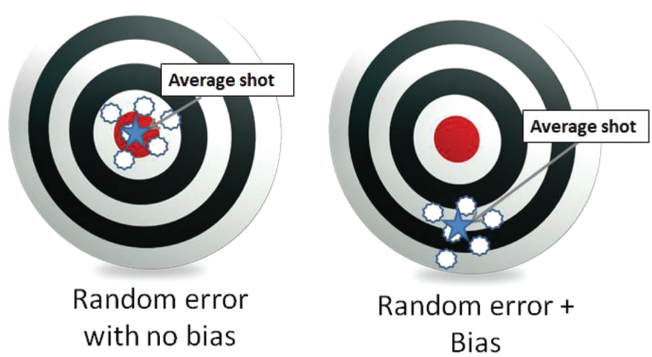

Figure 5. Illustration of random error and errors due to bias.

stretched out, the measures then taken will consistently underestimate its true height. And aggregating a large number of measures will not eliminate bias. These concepts are illustrated in Figure 5.

Though simplistic examples of measuring a desk or flipping a coin are used here to explain the concepts the same precepts apply to more complicated analytic tasks. While other reasons cited for why aggregated analyses do well are available, the focus here has been on the elimination of random error.[25]

AGGREGATING OPINIONS IS NOT NATURAL

In situations where multiple forecasting opinions have been made available on a given topic, identifying which individual forecaster is best and most accurate seems appropriate. Once the determination is made, relying more heavily on that forecaster in the future and discarding the rest seems logical.[26] Similarly, for intelligence problems, that the best or most experienced analyst on the topic determining the Intelligence Community's assessment seems natural. But aggregation suggests something different. Rather than discarding or ignoring the input of others, a large body of research spanning over the past 100 years has indicated that better accuracy can be achieved through the aggregation of a large number of opinions.

In the examples given here, opinions were aggregated simply by averaging. Using this approach, the opinion of each individual counts equally. But should this be the case? Some individuals may have demonstrated better accuracy in the past, or have more experience and should therefore have their opinions weighted more heavily. In the smile video task, for example, I found one variable that did predict performance. Participants were asked, before they tried the task, to estimate how many videos they thought they would get correct. Significantly, participants' estimates were positively correlated to the number of videos they actually did get correct.[27] People may also have some intuition of how well they can do a task. Thus using one's own estimate of performance as a way to weight some opinions more heavily while forming the aggregate assessment might be possible. What is the optimal way to weigh the opinions? The IARPA's ACE project is exploring such possibilities.

REFERENCES

[1] The views expressed in the article are those of the author and do not necessarily reflect the official policy or position of the U.S. Government, the Department of Defense, or any of its components.

[2] For a review of the brainstorming literature, see Brian Mullen, Craig Johnson, and Eduardo Salas, ''Productivity Loss in Brainstorming Groups: A Meta-Analytic Integration,'' *Basic and Applied Social Psychology,* Vol. 12, No. 1, 1991, pp. 3–23.

[3] James Surowiecki, *The Wisdom of Crowds* (New York: Anchor Books, 2005). Also see Louise S., ''The Swarm Is Greater Than the Parts: Cultivating the Wisdom of Crowds in the Intelligence Community,'' 2006 Galileo Paper (2006).

[4] Francis Galton, ''The Ballot-Box,'' *Nature,* Vol. 75, No. 1952, 1907, p. 509. Francis Galton, ''Vox Populi,'' *Nature,* Vol. 75, No. 1949, 1907, pp. 450–451.

[5] Alison Hubbard Ashton, ''An Empirical Study of Budget-Related Predictions of Corporate Executives,'' *Journal of Accounting Research,* Vol. 20, No. 2, 1982, pp. 440–449. Alison Hubbard Ashton and Robert H. Ashton, ''Aggregating Subjective Forecasts: Some Empirical Results,'' *Management Science,* Vol. 31, 1985, pp. 1499–1508.

[6] J. M. Bates and C. W. J. Granger, ''The Combination of Forecasts,'' *OR,* 1969, 451–468.

[7] Martin Spann and Bernd Skiera, ''Internet-Based Virtual Stock Markets for Business Forecasting,'' *Management Science,* Vol. 49, No. 10, 2003, pp. 1310–1326.

[8] ''IARPA-ACE Program,'' accessed 31 January 2013, at http://www.iarpa.gov/Programs/ia/ACE/ace.html

[9] K. J. Arrow, R. Forsythe, M. Gorham, R. Hahn, R. Hanson, J. O. Ledyard, S. Levmore, et al. ''The Promise of Prediction Markets,'' *Science,* Vol. 320, No. 5878, 2008, p. 877.

[10] InTrade recently had to cease trading due to U.S. Internet gambling laws. Steven Russolillo, ''Intrade Shuts Down Months After Federal Scrutiny—MarketBeat—WSJ,'' The *Wall Street Journal*, 11 March 2013, at http://blogs.wsj.com/marketbeat/2013/03/11/intrade-shuts-down-months-after-federal-scrutiny/

[11] James E. Kajdasz, ''Spot the Fake Smile,'' at http://www.surveymonkey.com/s/SmileRead, accessed 13 May 2013. The test was modified from the original: ''BBC—Science & Nature—Human Body and Mind—Spot The Fake Smile,'' http://www.bbc.co.uk/science/humanbody/mind/surveys/smiles/, accessed 13 May 2013.

[12] One-Sample $t(216) = 19.62$, $p < .001$.

[13] $M = 13.59$, $SD = 2.71$, $CI_{.90}$: 9.15 to 18.03.

[14] $M = 13.59$, $SE = 0.59$, $CI_{.90}$: 12.62 to 14.56.

[15] $M = 17.2$, $SD = 1.45$, $CI_{.90}$: 14.81 to 19.58.

[16] Independent Samples $t(273.47) = 34.06$, $p < .001$ (adjusted for unequal variances, according to Levine's test).

[17] $M = 15.03$, $SD = 2.21$.

[18] $M = 17.42$, $SD = 1.24$.

[19] Pearson's $r = .37$, $p < .001$.

[20] T. H. Ho and K. Y. Chen, ''Discovering and Managing New Product Blockbusters: The Magic and Science of Prediction Markets,'' *California Management Review,* No. 50, 2007, pp. 144–158.

[21] $M = 13.59$, $SD = 2.71$, $CI_{.90}$: 9.15 to 18.03.

[22] $M = 17.2$, $SD = 1.45$, $CI_{.90}$: 14.81 to 19.58.

[23] $M = 17.42$, $SD = 1.24$, $CI_{.90}$: 15.39 to 19.45.

[24] K. J. Arrow, R. Forsythe, M. Gorham, R. Hahn, R. Hanson, J. O. Ledyard, S. Levmore, et al. ''The Promise of Prediction Markets,'' p. 877. Alison Hubbard Ashton and Robert H. Ashton, ''Aggregating Subjective Forecasts: Some Empirical Results,'' pp. 1499–1508. J. M. Bates and C. W. J. Granger. ''The Combination of Forecasts,'' pp. 451–468. Robin M. Hogarth, ''A Note on Aggregating Opinions,'' *Organizational Behavior and Human Decision Processes,* Vol. 21, No. 1, 1978, pp. 40–46. Martin Spann and Bernd Skiera, ''Internet-based Virtual Stock Markets for Business Forecasting,'' pp. 1310–1326. G. Tziralis and I. Tatsiopoulos, ''Prediction Markets: An Extended Literature Review,'' *The Journal of Prediction Markets,* Vol. 1, No. 1, 2007, pp. 75–91. Justin Wolfers and Eric Zitzewitz, ''Prediction Markets,'' *The Journal of Economic Perspectives,* Vol. 18, No. 2, Spring 2004, pp. 107–126. Puong Fei Yeh, ''Using Prediction Markets to Enhance US Intelligence Capabilities,'' *Studies in Intelligence,* Vol. 50, No. 4, 2006, pp. 37–49.

[25] James Surowiecki notes for example: it increases the diversity of opinions, estimates cannot be dominated by ''bossy'' group members, and forecaster specialization allows members to bring their unique knowledge to the problem.

[26] J. M. Bates and C. W. J. Granger, ''The Combination of Forecasts,'' *OR,* 1969, pp. 451–468.

[27] Pearson's $r = .21$, $p = .002$.